JOMHACK
EMPOWERING INNOVATION

```
> bootcamp.learn()
```

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JomCode Coding Bootcamp

CHAPTER 01

Development environment

```
follow / verify / explain
```

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The tools and the goal

01 Python runs the backend. Node.js and npm run/build frontend tools.

02 Git records changes. GitHub hosts repositories. VS Code edits files and provides a terminal.

03 Complete each installation, reopen the terminal and verify its version.

04 Use your operating system section. Installer appearance and version labels can change.

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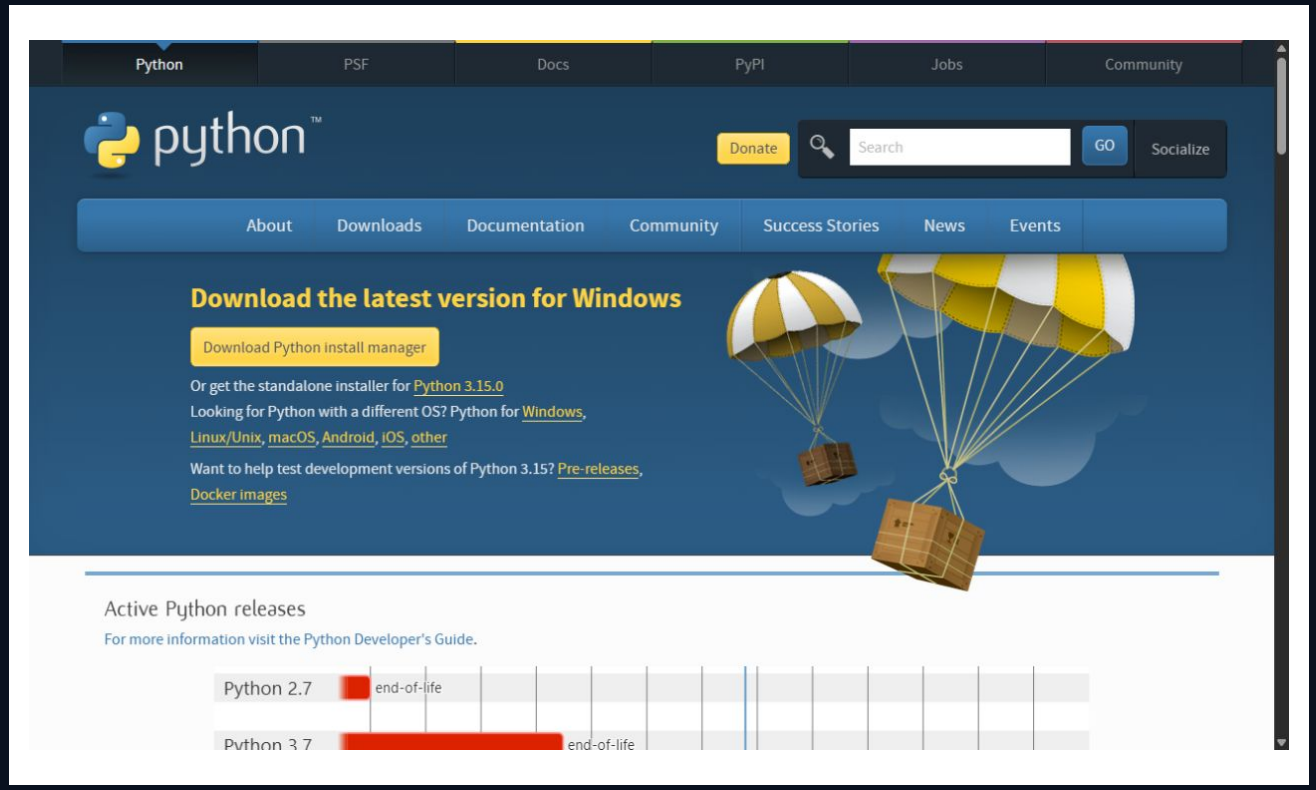
Python download

1. Open python.org/downloads. Use a supported Python 3 release with packages available for it.

2. For this course, prefer Python 3.14 or 3.13 over a newly released interpreter until dependency installation succeeds.

3. Select Windows or macOS. Choose a standard installer for your OS and CPU, avoiding source archives and prereleases.

4. Windows: select the .exe installer. macOS: select the universal2 .pkg installer.



Official screen reference; interface labels may change.

Official reference

Python installation on Windows

1. Open Downloads and run the Python installer obtained from python.org.

2. For the traditional .exe installer, select Add python.exe to PATH. Keep the Python launcher enabled.

3. Select Install Now. Approve the normal installer prompt if shown, then wait for completion.

4. Close the installer. Close existing terminals and open a new PowerShell window.

5. Run the checks below. If py is unavailable, try python and inspect the installation/PATH before continuing.

```
py --version
py -m pip --version
```

Official reference

Python installation on macOS

1. Open the downloaded python.org .pkg file.

2. Follow Introduction, License and Installation prompts. Choose the normal installation destination.

3. Complete the installer and close it. If supplied, run Install Certificates.command in the installed Python folder.

4. Open a new Terminal window and run the checks below. Use python3 for macOS commands throughout this course.

5. Confirm the reported interpreter matches the installation you intend to use.

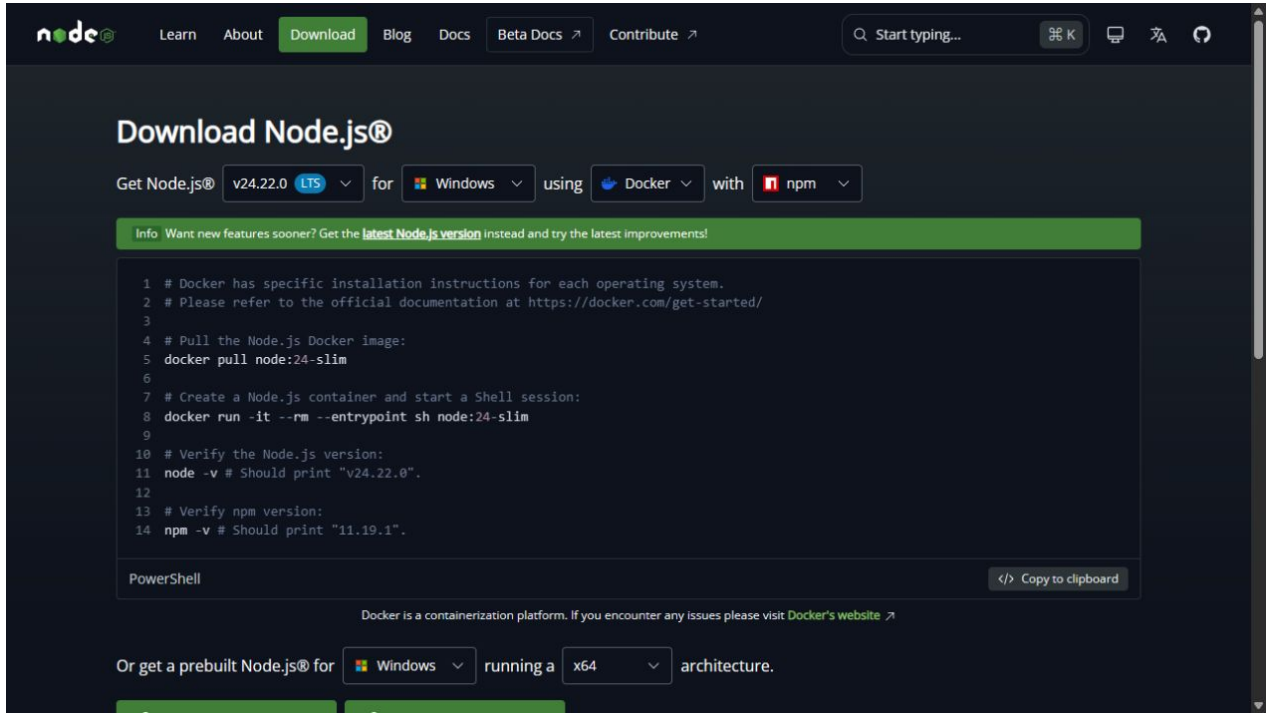
```
python3 --version
python3 -m pip --version
```

Official reference

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Node.js download

1. Open nodejs.org/en/download.
2. Select the LTS release. Select your OS and CPU architecture.
3. Choose the prebuilt installer: Windows .msi or macOS .pkg. The course does not require Docker.
4. The project needs Node 20.19+ or 22.12+; a compatible current LTS is preferable.



Official screen reference; interface labels may change.

Official reference

Node.js installation and checks

Windows: open the .msi, proceed through the wizard and keep npm and Add to PATH enabled.

macOS: open the .pkg, follow the installer and keep the standard Node/npm installation.

Finish installation, close existing terminals and open a new one.

Run both checks. Node executes JavaScript; npm installs the frontend dependencies.

If PowerShell blocks npm.ps1, use npm.cmd for npm commands instead of changing machine security settings.

```
node --version
npm --version
```

Official reference

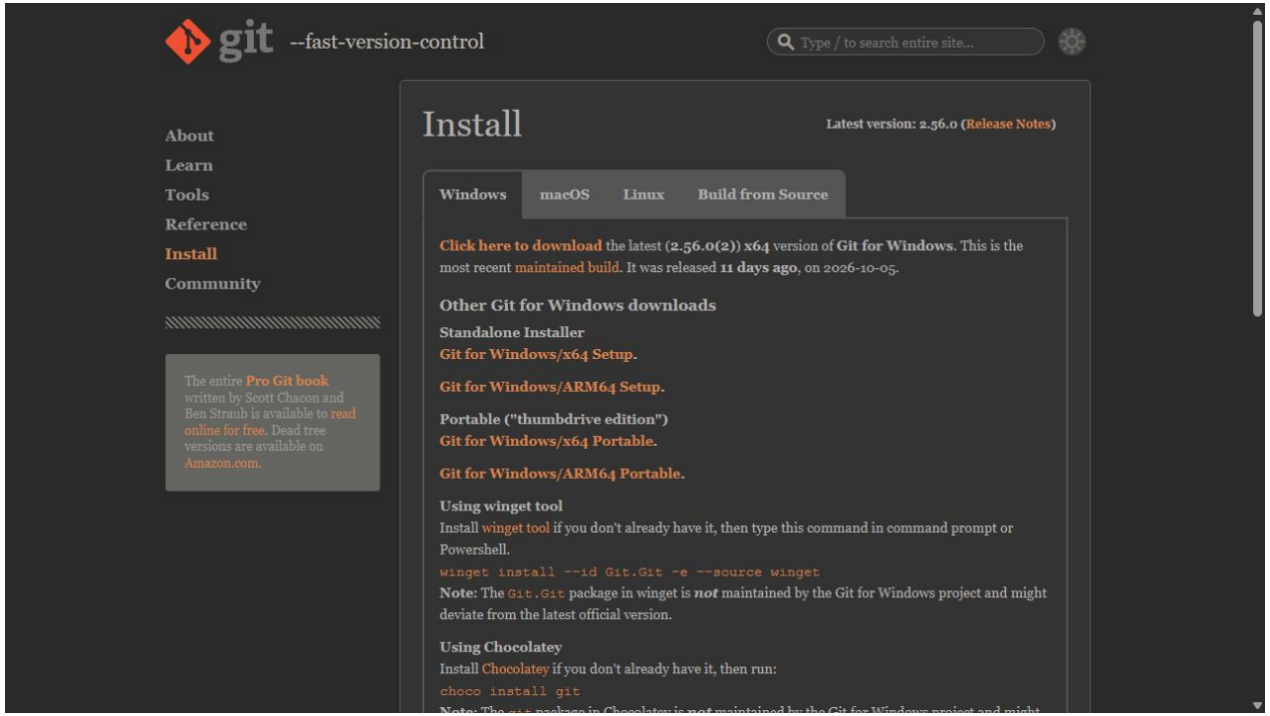
Git download

1. Check git --version first. If a version appears, Git is already available.

2. Windows: open git-scm.com/install/windows and choose the matching Git for Windows installer.

3. macOS: running git --version can offer Apple Command Line Tools. Install them, or follow the official macOS instructions.

4. VS Code uses an installed Git executable; installing VS Code alone does not install Git.



Official screen reference; interface labels may change.

Official reference

Git installation and checks

Windows: run the downloaded installer and keep recommended defaults.

At the PATH choice, retain Git from the command line and also from third-party software.

Complete the wizard. No Unix-only shell setup is needed for the PowerShell examples.

macOS: finish the Command Line Tools installation if prompted.

Open a new terminal, then run git --version. A command-not-found message means installation/PATH needs attention.

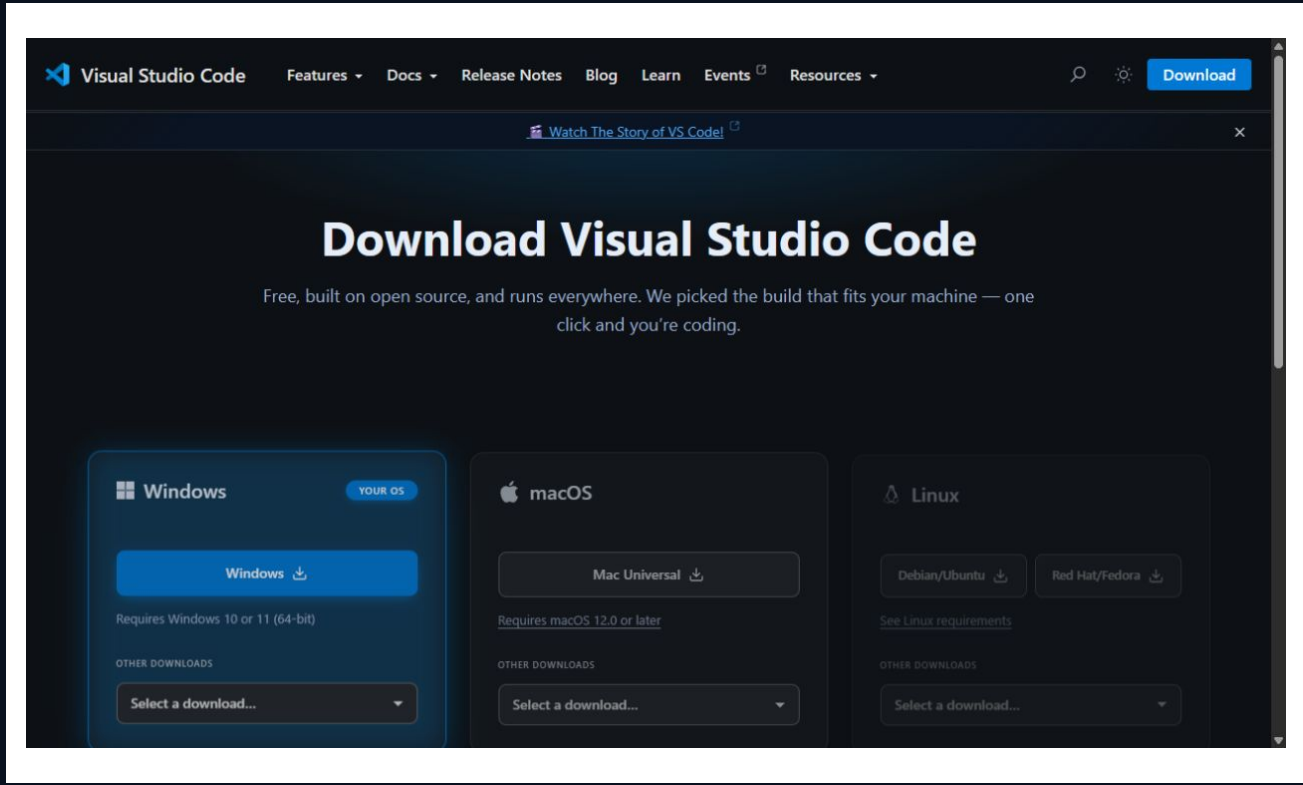
```
git --version
```

Official reference

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VS Code download

1. Open code.visualstudio.com/download.
2. Windows: choose User Installer for your CPU. macOS: choose the matching Apple silicon or Intel download.
3. Download the stable release from the official website.
4. Keep the screenshot as a location reference; button labels can change.



Official screen reference; interface labels may change.

Official reference

VS Code installation on Windows

1. Run the downloaded User Installer and follow its prompts.

2. Keep Add to PATH enabled. Open with Code context-menu options are useful but optional.

3. Finish the installation and open VS Code.

4. Close/reopen any terminal, then check code --version.

5. In VS Code Extensions, install Python published by Microsoft. Open your practice folder through File > Open Folder.

```
code --version
```

Official reference

VS Code installation on macOS

1. Open the downloaded archive if it has not expanded automatically.
2. Drag Visual Studio Code.app into Applications, then open it.
3. Open the Command Palette with Shift+Command+P.
4. Run Shell Command: Install code command in PATH.
5. Open a new Terminal, run code --version, then install the Microsoft Python extension inside VS Code.

```
code --version
```

Official reference

Terminal and file navigation

VS Code: Terminal > New Terminal opens a shell in the current workspace.

pwd shows the current folder. cd changes folder. Quote paths containing spaces.

Windows PowerShell: Get-ChildItem lists files. macOS Terminal: ls lists files.

Open a folder named jomcode-practice. Save files before running commands.

Run one command per line. Do not type printed prompt symbols or OS labels.

```
pwd
cd "PATH_TO_YOUR_PRACTICE_FOLDER"
```

First saved Python file

1. In VS Code, open jomcode-practice and create hello.py.

2. Type print("Hello, JomCode") and save.

3. Open the integrated terminal in that folder.

4. Windows: py hello.py. macOS: python3 hello.py.

Expected result: Hello, JomCode. A missing-file error usually means an incorrect working folder or filename.

```
print("Hello, JomCode")
```

Final verification on Windows

Run every command in a fresh terminal. Each must print a version or interpreter/package path.

Use npm.cmd --version if the shell blocks npm.ps1.

Save a screenshot or record the version output as your setup evidence.

```
py --version
py -m pip --version
node --version
npm --version
git --version
code --version
py hello.py
```

Final verification on macOS

Run every command in a fresh Terminal. Each must print a version or interpreter/package path.

If code is missing, repeat the VS Code Shell Command step.

Continue only when the tools work and hello.py prints the expected greeting.

```
python3 --version
python3 -m pip --version
node --version
npm --version
git --version
code --version
python3 hello.py
```